

Lesson 13

■ User Experience Design

Slide Set to accompany

Software Engineering: A Practitioner's Approach, 8/e

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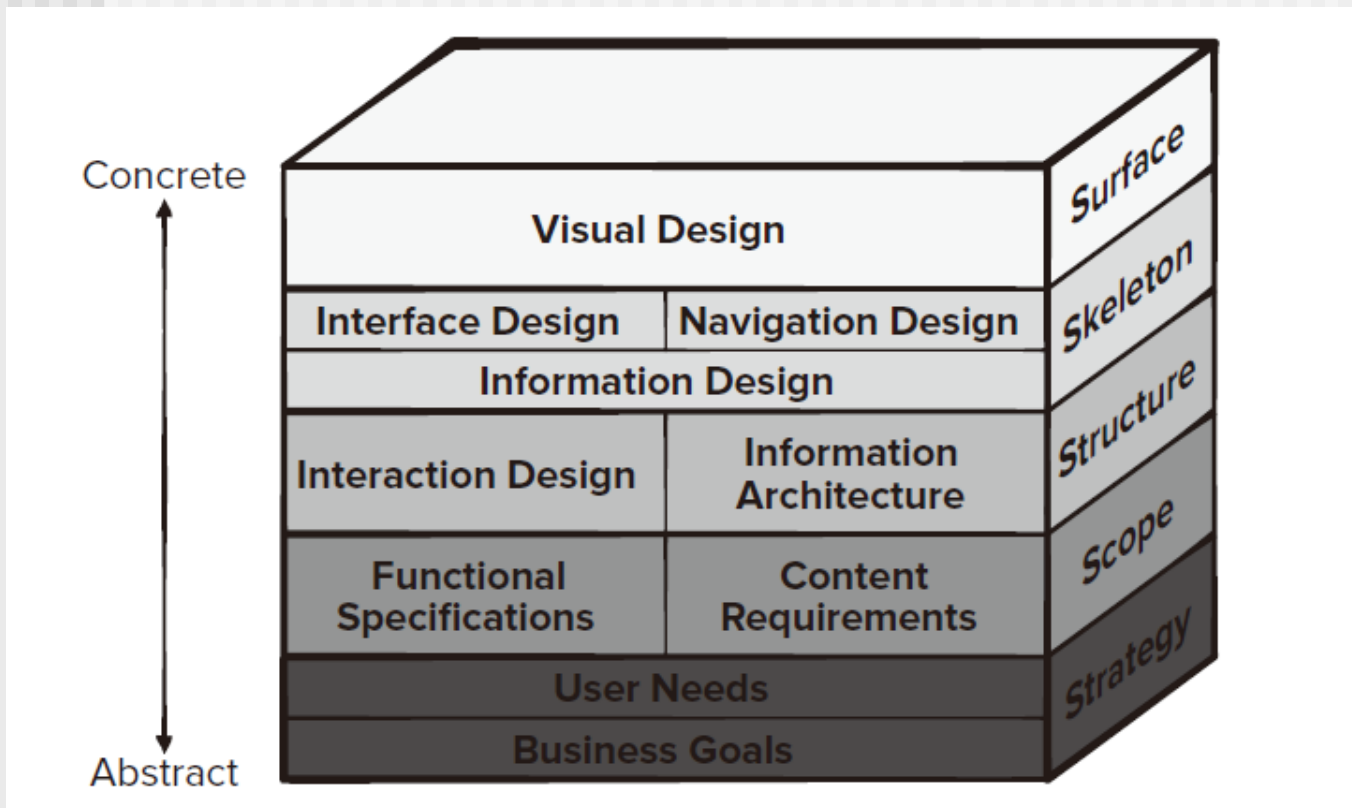
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User Experience Design Elements



Golden Rules

- Place the **user** in control.
- Reduce the **user's** memory load.
- Make the **interface** consistent.

Place the User in Control

- Define interaction modes in a way that does not force a user into unnecessary or undesired actions.
- Provide for flexible interaction.
- Allow user interaction to be interruptible and undoable.
- Streamline interaction as skill levels advance and allow the interaction to be customized.
- Hide technical internals from the casual user.
- Design for direct interaction with objects that appear on the screen.

Reduce the User's Memory Load

- Reduce demand on short-term memory.
- Establish meaningful defaults.
- Define shortcuts that are intuitive.
- The visual layout of the interface should be based on a real world metaphor.
- Disclose information in a progressive fashion.

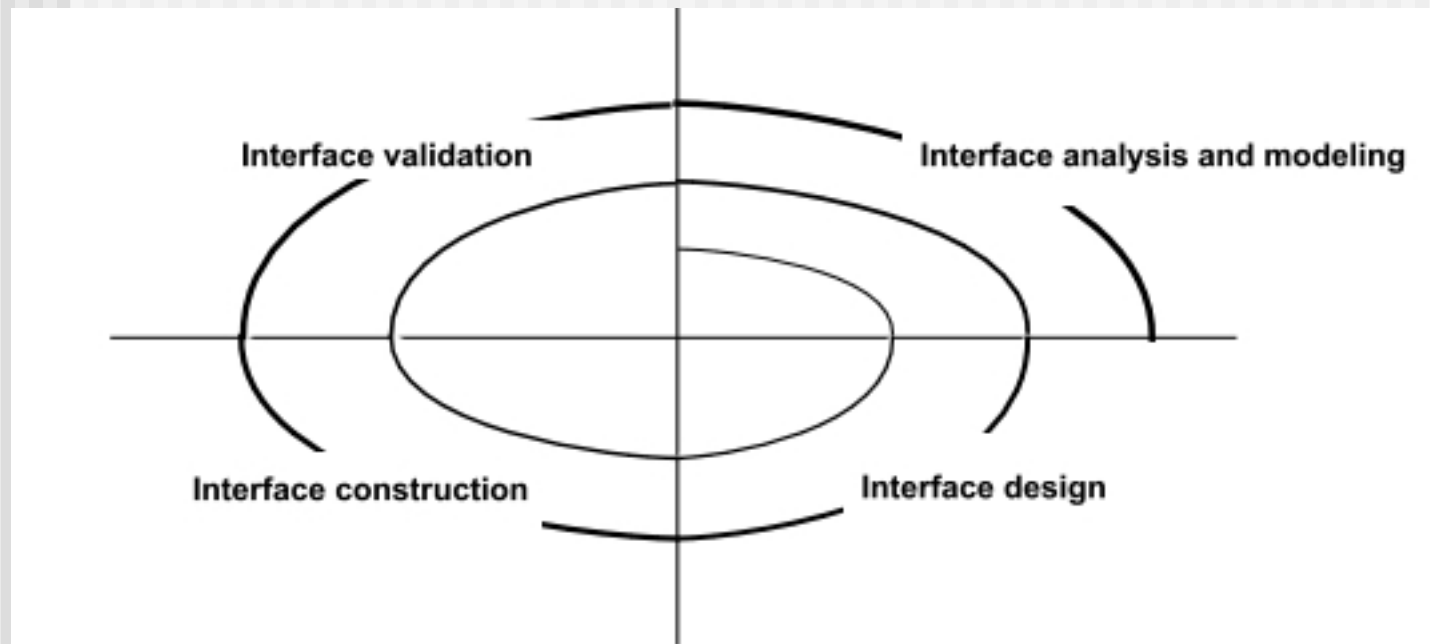
Make the Interface Consistent

- Allow the user to put the current task into a meaningful context.
- Maintain consistency across a family of applications.
- If past interactive models have created user expectations, do not make changes unless there is a compelling reason to do.

User Interface Design Models

- **User model** — a profile of all end users of the system.
- **Design model** — a design realization of the user model.
- **Mental model (system perception)** — the user's mental image of what the interface is.
- **Implementation model** — the interface “look and feel” coupled with supporting information that describe interface syntax and semantics.

User Interface Design Process



Interface Analysis

- Interface analysis means understanding
 - the people (end-users) who will interact with the system through the interface;
 - the tasks that end-users must perform to do their work,
 - the content that is presented as part of the interface
 - the environment in which these tasks will be conducted.

User Analysis

- Are users trained professionals, technician, clerical, or manufacturing workers?
- What level of formal education does the average user have?
- Are the users capable of learning from written materials or have they expressed a desire for classroom training?
- Are users expert typists or keyboard phobic?
- What is the age range of the user community?
- Will the users be represented predominately by one gender?
- How are users compensated for the work they perform?
- Do users work normal office hours or do they work until the job is done?
- Is the software to be an integral part of the work users do or will it be used only occasionally?
- What is the primary spoken language among users?
- What are the consequences if a user makes a mistake using the system?
- Are users experts in the subject matter that is addressed by the system?
- Do users want to know about the technology the sits behind the interface?

Task Analysis and Modeling

- Answers the following questions ...
 - What work will the user perform in specific circumstances?
 - What tasks and subtasks will be performed as the user does the work?
 - What specific problem domain objects will the user manipulate as work is performed?
 - What is the sequence of work tasks—the workflow?
 - What is the hierarchy of tasks?
- **Use-cases** define basic interaction
- **Task elaboration** refines interactive tasks
- **Object elaboration** identifies interface objects (classes)
- **Workflow analysis** defines how a work process is completed when several people (and roles) are involved

Analysis of Display Content

- Are different types of data assigned to consistent geographic locations on the screen (e.g., photos always appear in the upper right hand corner)?
- Can the user customize the screen location for content?
- Is proper on-screen identification assigned to all content?
- If a large report is to be presented, how should it be partitioned for ease of understanding?
- Will mechanisms be available for moving directly to summary information for large collections of data.
- Will graphical output be scaled to fit within the bounds of the display device that is used?
- How will color to be used to enhance understanding?
- How will error messages and warning be presented to the user?

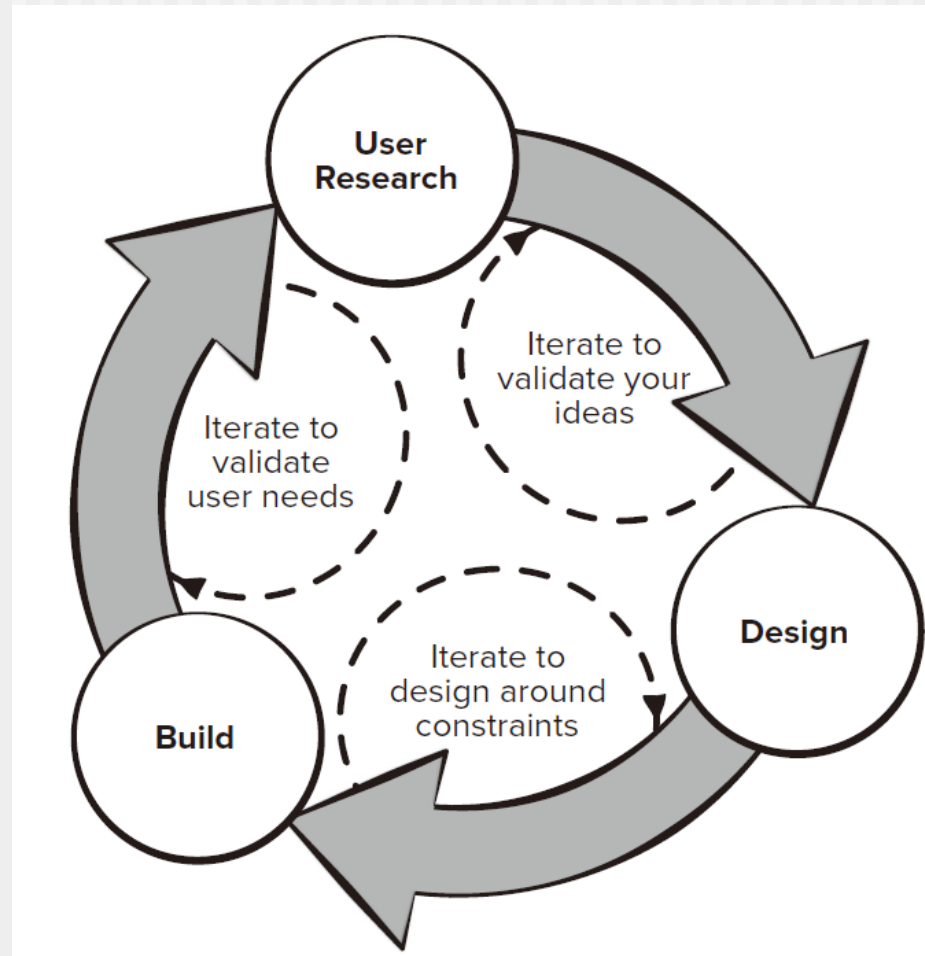
Interface Design Steps

- Using information developed during interface analysis, **define interface objects and actions (operations)**.
- **Define events (user actions)** that will cause the state of the user interface to change. Model this behavior.
- **Depict each interface state** as it will actually look to the end-user.
- **Indicate how the user interprets the state of the system** from information provided through the interface.

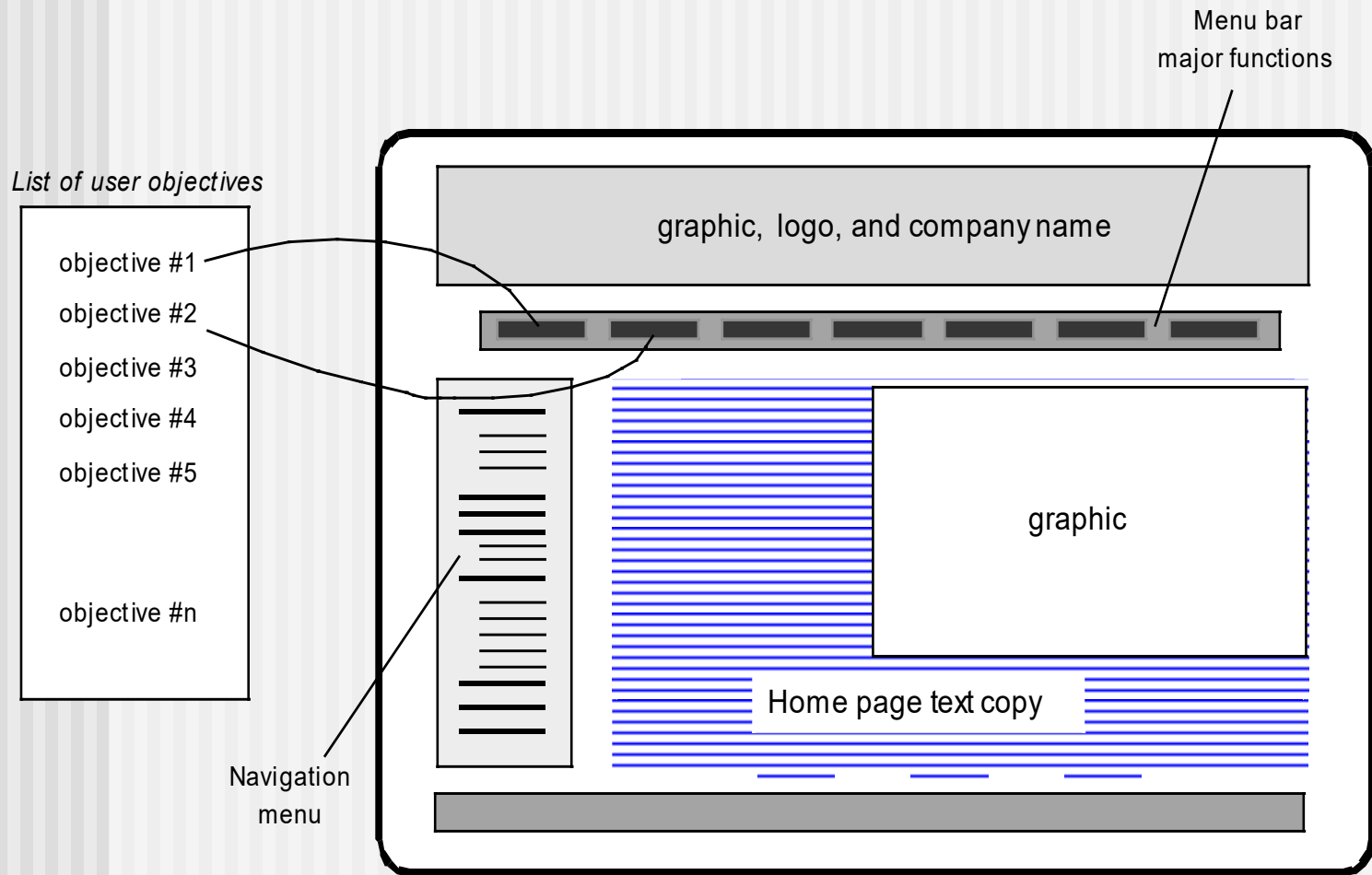
Design Issues

- Response time
- Help facilities
- Error handling
- Menu and command labeling
- Application accessibility
- Internationalization

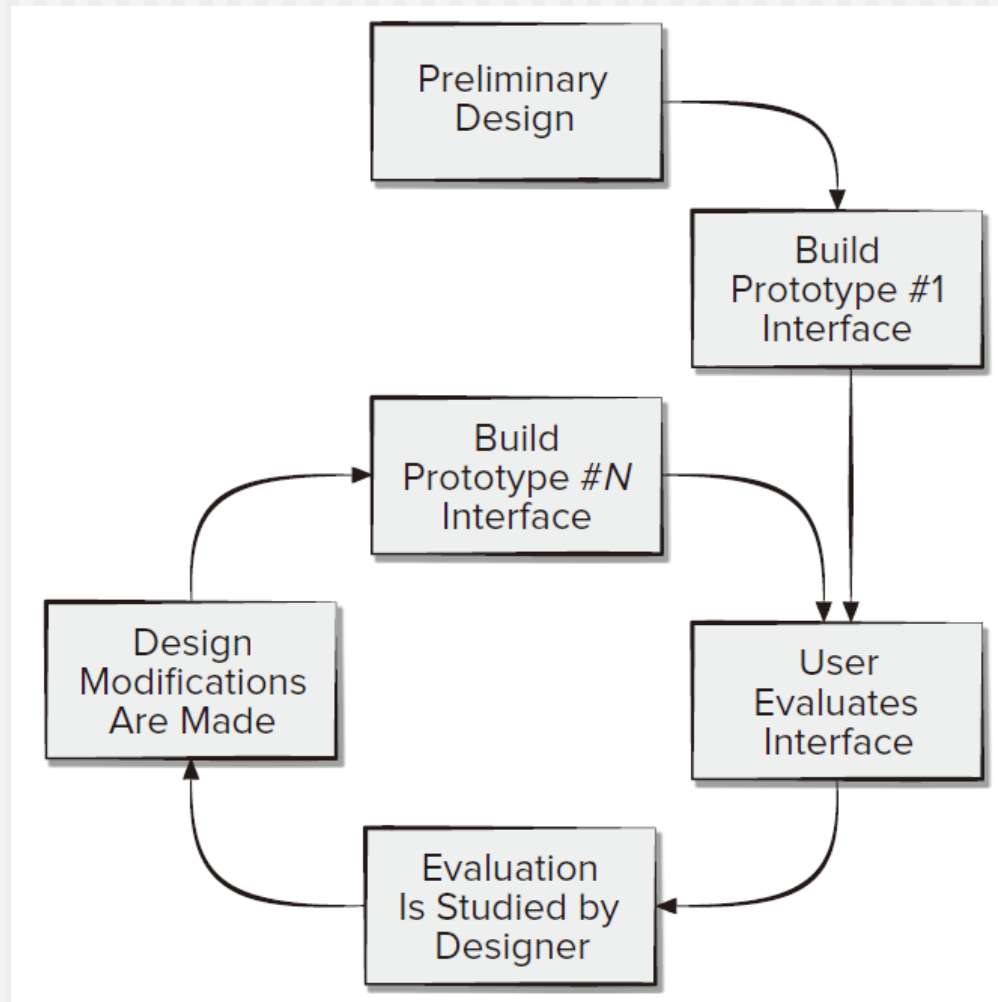
User Experience Design



User Interface Design



Design Evaluation



Usability & Accessibility

- *Where am I?* The interface should
 - provide an indication of the Web or Mobile App that has been accessed
 - inform the user of her location in the content hierarchy.
- *What can I do now?* The interface should always help the user understand his current options
 - what functions are available?
 - what links are live?
 - what content is relevant?
- *Where have I been, where am I going?* The interface must facilitate navigation.
 - Provide a “map” (implemented in a way that is easy to understand) of where the user has been and what paths may be taken to move elsewhere within the Web or Mobile App.

Assignment

- Review for the exam

 - 《Software Engineering》 (8th Edition)

 - Chapter 1 to Chapter 18

 - 《Software Engineering》 (9th Edition)

 - Chapter 1 to Chapter 12